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Gentrification and Displacement: Modeling a Complex Urban Process

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ABSTRACT

To shed some light on longstanding questions around gentrification, in this research we model environmental gentrification and gentrification-related displacement of residents. We do this through the development of an agent-based model of a simple urban region, considering different urban contexts and policy approaches to polluted facilities and the relationship of these policies with subsequent gentrification and displacement. We find that gentrification-related displacement is most likely, and most impactful, in urban regions characterized by high levels of density and low levels of residential segregation preferences. Displacement is far less prevalent in low-density regions, particularly those with high segregation preferences. We discuss the potential for different policy implications in these different urban contexts.

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Gentrification is a difficult concept to define, to study, and to craft policies for (Rayle, 2015). The term is ideologically laden (Freeman, 2011), it is difficult to conceptually separate and measure the various aspects of the gentrification process (Slater, 2006), and the line between economic development and gentrification is sufficiently blurred that crafting policies to address the issue is a challenge (Clark, 2005).

In general terms, gentrification is thought to occur when neighborhoods with lower socioeconomic status (SES) transition to higher SES (Hamnett, 1991). This implies a two-part process—the movement in of higher SES residents, and the moving out of lower SES residents (Slater, 2006). This inflow and outflow of residents can be considered part of the same process, as the Centre for Urban Studies (1964) conceived of gentrification when it coined the term, but they may also be fundamentally different processes. Further, it is the flow out of lower SES residents that are considered to be potentially problematic, for these may be involuntary and harmful to lower SES residents, but an inflow of higher SES residents that does not displace lower SES residents may have economic benefits. Our approach in this article is to disentangle this complex issue through the use of a simulation model that allows us to explore the in-migration of high-SES residents separately from the out-migration of lower SES residents.

Specifically, we consider the nature of the relationship between the in-migration process, which we will refer to as *gentrification*, and the out-migration process, which we will refer to as *displacement*, or gentrification-related displacement. In doing so, we address challenges of assessing and measuring gentrification and displacement (Grossmann & Haase, 2016) through the use of an agent-based simulation model (ABM) of an urban system that enables us to track the movement of city residents over time. Although the model is a simulation of a city, it allows us to explore the possibilities for and pressures on different types of residents as a city grows and changes over time.

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We track where gentrification and displacement take place, and also the burdens that are placed on residents in neighborhoods that gentrify, specifically assessing the extent to which displacement of existing residents occurs and what conditions exacerbate or ameliorate this outcome (Eckerd & Reames, 2012).

In the sections that follow, we describe challenges of studying gentrification and displacement, and how gentrification is conceived in the literature, and then describe our approach and model. The simulation results follow, and we end with some implications for housing policy.

Gentrification

Gentrification has been defined in a variety of ways, ranging from benign processes of housing market sorting (Smith, 1979) to militaristic metaphors of invasion and conquest (Hamnett, 1991). At root, gentrification is a process of neighborhood change (Grossmann & Haase, 2016). To gentrify, a neighborhood is first populated primarily by lower wealth residents and subsequently populated by higher wealth residents. Whereas some view the process as a fundamentally class-based phenomenon (Centre for Urban Studies, 1964), others call attention to racial components as white populations increase in neighborhoods that had previously been primarily African American or Hispanic (at least in the American context; Hwang & Sampson, 2014; Schaffer & Smith, 1986). Some argue that gentrification only occurs if lower wealth residents are displaced (LeGates & Hartman, 1986), whereas others contend that gentrification and displacement are conceptually distinct (Slater, 2006).

Here we take the latter perspective, for there are wide-ranging implications of gentrification and, despite the general tone of much of the literature, not all are necessarily negative (Freeman, 2008). Although reinvestment may displace people, it can also bring more jobs, services, and safety to neighborhoods that have lacked these needs (Bartik, 1991; Valler & Wood, 2010). Some residents can benefit financially through higher property values and deconcentration of poverty (Sullivan, 2006). Of course there are possible negative implications, too. Residents may well lose their sense of what their neighborhood was and who it was for (Freeman, 2011), or end up worse off than they were before if rents rise and they are forced to move into less appealing neighborhoods and also bear the burdens of moving (Vigdor, 2002).

If reinvestment benefits both existing and incoming residents, we may want policy to encourage more redevelopment, but if existing residents are burdened with displacement, then gentrification may need to be tempered (Clark, 2005). That is, if little displacement occurs, the concept of gentrification may not be fundamentally different from economic development (Sullivan, 2007), but if much displacement occurs, then the implications of this process could be burdening some of the very people that economic development is usually intended to benefit.

This is all complicated by the fact that we do not know the scale of the displacement problem (Slater, 2006). We certainly have evidence that displacement occurs in certain areas (Lyons, 1996; Vigdor, 2002), but there is little generalizable evidence regarding just how much displacement occurs (Hwang, 2016). Schill and Nathan (1983) and Vigdor (2002) note that displacement, although a key factor in considerations about the implications of gentrification, can be very difficult to assess. It is problematic to differentiate residents who were displaced from those who would have relocated anyway. This may be because we only know that gentrification has taken place after the fact, or at least once the process begins. This means that we cannot readily identify areas that *will* gentrify, only those that either have undergone or are currently undergoing gentrification, and so potential displacees cannot be identified and tracked prior to the gentrification. This is particularly problematic because the first displacees may well be the only people displaced, and are the most vulnerable assuming that the first movers are those least able to afford rent increases (Clay, 1979; Fusch, 1980; Hammel & Wyly, 1996; Kennedy & Leonard, 2001; Ley, 1981; Smith, 1979; Wyly & Hammel, 1998).

Tracking the movement of individual residents is difficult to do on a scale large enough to provide generalizable insight—it is easier to assess who has moved into a geographic area than to assess the disparate movements of people away from that area (Vigdor, 2002). Further, even if residential movement could be tracked systematically, it would be difficult to separate the circumstances under which individuals left gentrifying communities. Some may have been displaced because of higher rents, but others may have relocated for work, and others might have moved regardless of gentrification, whereas others still may have been taking advantage of increases in their home's value to sell in a rising real estate market (Rayle, 2015).

Further, gentrification processes take place in the broader context of urban residential sorting patterns (Sanchez & Dawkins, 2001), and are affected by land uses (Wyly & Hammel, 1998), and economic considerations such as commuting costs (Brueckner & Rosenthal, 2009)—themselves affected by the transportation infrastructure. They are affected by the housing stock (Smith, 1979) and living preferences of the regional population (Ley, 1986), so identifying when gentrification and gentrification-related displacement occur is challenging.

Some types of cities appear to be more prone to gentrification than others. Large cities, especially those with rapid job growth and tight housing markets, seem more prone to gentrification, as do cities with high traffic congestion (e.g., inefficient transportation infrastructure), and public policies (notably tax breaks) that encourage people to live in redeveloping neighborhoods (Black, 1980; Kennedy & Leonard, 2001; Rose, 1984; Wyly & Hammel, 1998). Gentrification may be more prone to occur in cities with high transportation costs, but also in those with good access to public transit. It may also be tied to the design of the urban setting, in particular the architecture of available buildings and the aesthetic nature of the community (Eckerd, 2011; Kennedy & Leonard, 2001; Kolko, 2007). Gentrification may also be more likely in areas where residents prefer, or at least accept, higher levels of density—sometimes because of geographic features, general residential preferences, or even policies intended to curb sprawl (Dieleman & Wegener, 2004).

Other public policies have been used as economic development drivers that have also been tied to gentrification. One hallmark of recent urban redevelopment is the remediation of environmental hazards, transforming polluting disamenities into new urban amenities, like parks, mixed-use development projects, or creative spaces. In certain locations, these efforts have been tied to subsequent gentrification (Checker, 2011). This transition from disamenties to amenities can create incentives for investment in a community that would not have previously appealed to outsiders because of the disamenity (Eckerd, 2011). However, this process is difficult to observe as well, because not all regions that receive improvement will gentrify.

In short, there are complex, interweaving systems of urban policy, community activism, residential taste, transportation systems, and real estate market dynamics that foster, but do not promise, gentrification in specific locations. Empirically, it is difficult to know when and how gentrification may occur, if and which residents will be displaced, and the implications of this displacement. Additionally, there are challenges because studies show that different ways of measuring gentrification, in terms of demographic outcomes, the changes in amenities, and the areal unit of analysis one uses, often lead to different conclusions (Eckerd, 2011).

When we consider the many factors and interactions that go into determining whether gentrification occurs and its consequences for displacement, it becomes apparent that gentrification and displacement occur as emergent outcomes of a complex system. A complex system is characterized by a set of interacting components, such as economic systems or social systems, where the aggregate result of these interactions is a nonlinear, emergent response (Holland, 1998; Miller & Page, 2007). In simpler terms, the emergent occurrences of a complex system are different and more than the sum of the system's parts. When complexity is a key feature of system dynamics, simulation methods, such as ABM, can be a useful approach to further understanding (Epstein, 2006). Rather than aiming to differentiate the independent effects of some set of independent variables on some outcome variable, simulation methods help observe the behavior of a system that emerges from interdependent inner working mechanisms. Therefore, the results of any

particular simulation trial are the consequence of an interacting set of individual actors within a dynamic setting. The assumption is not that there are independent effects that lead to some outcome, but rather that systemic social patterns emerge from a distinct *recipe* of the interaction of different social systems. Other scholars have used computational simulation methods to understand related issues (Epstein, 2006).

The utility of ABMs for urban analysis is well established (Batty, 2007). ABMs have been used to learn more about urban development and sprawl (Brown & Robinson, 2006), emergency management strategies (Chen & Zhan, 2008), environmental justice (Campbell et al., 2015), and land use (Filatova, Parker, & Van der Veen, 2009). Complexity analysis techniques such as computational simulations have been used in studies of traffic (Nagel & Rickert, 2001) and land-use (Engelen, White, Uljee, & Drazan, 1995). Beauregard (1986) was perhaps the first to explicitly connect the notions of complexity and gentrification, and since then several models have assessed the issue. O'Sullivan's (2002) model suggests that as real estate value declines and reaches its nadir, property values in neighborhoods with amenities can suddenly and rapidly increase, such that a neighborhood can fully gentrify within just a few years. Torrens and Nara (2007) find that gentrifying neighborhoods may tend to remain distinct from other nearby areas, suggesting that the geographic limits of gentrification may be relatively small, perhaps defined by amenities, and that gentrification can be rather sporadic. Other related models aim to unpack the individual-level location-choice decision (Devisch, Timmermans, Arentze, & Borger, 2009), assess individuals' propensity to consider the local policy context (O'Sullivan, 2009), or consider how urban spatial constraints and economic pressures affect residential patterns (Haase, Lautenbach, & Seppelt, 2010), or how land-use change affects cities (Huang, Parker, Filatova, & Sun, 2014) or specific regions (Jackson, Forest, & Sengupta, 2008). Broadly speaking, this work, like much of the empirical work on gentrification, has tended to assess the systemic outcome of decision-making processes of incoming residents but not the scale of, or the burden placed on, displaced residents.

Our model is an attempt to broadly investigate gentrification and displacement. We fold in aspects of the things we know are associated with gentrification—proximity to amenities (and distance from disamenities), access to employment, an urban setting that tends to have more density than a suburban setting, and living in proximity to people similar to oneself (Eckerd, 2011; Helms, 2003; Kennedy & Leonard, 2001; Kolko, 2007)—and explore the potential for and the implications of displacement of the original residents.

The Agent-Based Model

As the above discussion underscores, gentrification and displacement are complex phenomena. They are not easily understood or predicted using conventional policy analysis, and are challenging to assess quantitatively, and although case studies have value, it is difficult to know whether the trends identified in particular cases provide a generalized understanding of the phenomenon. To address these challenges, we model this process using an agent-based computer simulation model, based on a real estate market across a set of different regional contexts.

Our approach with the model is parsimonious—meaning that we do not try to explicitly model the complexity of the real world, but rather to simplify a real urban system into a few key parts. We do so purposely to have a model that represents the complexity of a true urban system, but is also simple enough to draw meaningful conclusions from. The model is described in detail below, but, stated simply, the model is of an urban region consisting of heterogeneous residents with different characteristics such as race (majority or minority), wealth (drawn randomly from empirically based distributions), and living preferences, who select locations in a housing market affected by amenities and disamenities, residential movement, and density, quality, and price. Therefore, whereas some of the known factors associated with gentrification and displacement are explicitly contained in the model, some other factors noted above as important to gentrification and displacement are more implicitly contained in this structure. For example, a density preference

could be thought of as a corollary to the transportation infrastructure, since we would expect residents to exhibit a greater preference for density (*ceteris paribus*) when transportation is less efficient.

One advantage of the ABM is that we can track the movement of all of the agents throughout the entire simulation. We trace residential movement through time, noting where neighborhoods gentrify, how many residents are displaced because of gentrification, where residents who leave gentrified neighborhoods go, and how many times residents in certain demographic groups are displaced.

Model Structure

A graphical depiction of the model is provided in Figure 1. The urban region is modeled as a simple grid of 41 × 41 (1,681 total) plots. These plots are the locations that residents can choose as their *home* plots. In other words, the plots can be conceived of as land locations that residents (and *firms*) select. Some plots will be occupied and some will be vacant and thus open to new (residential or nonresidential) land uses, depending on the characterics of the setting, as described below. In some cases, modeling higher density preferences by residents, more than one *house* can be placed on a plot.[1] The size of the artificial city was calibrated to ensure that there was sufficient space for a large population of residents, but also that it was relatively confined enough to limit residential choice and to ensure that the simulation could compute efficiently on our available computing resources. In the simulation trials described below, we alter many of the characteristics of this *city* but we do not alter its geography from this grid. Plots have variable prices and quality measures, both of which affect the desirability of each plot to the residents. Price and quality are adjusted as described below. At the start of the simulation, the expected quality variable is set to 50 for all plots (and can ultimately range from 0 (the lowest quality) to 100 (the highest quality) as the simulation moves forward in time). Prices are drawn randomly from a gamma distribution centered at $173,000, with a standard deviation of $43,000, which are the median home value price

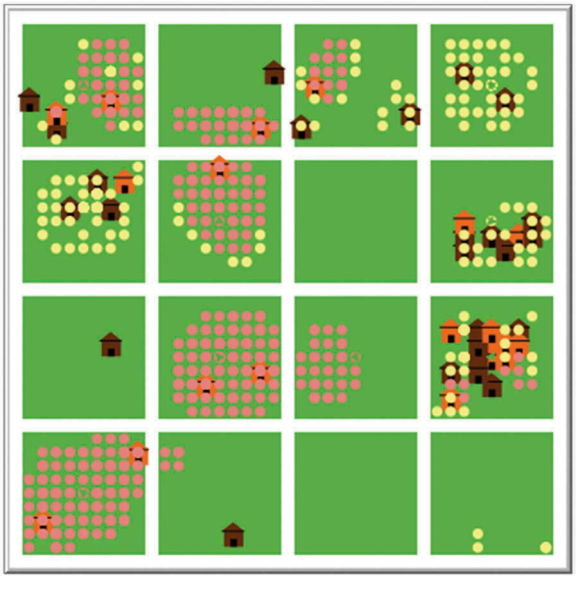


Figure 1. Graphic depiction of the urban region partway through one scenario.

Note. Dots represent residents (e.g., majorities or minorities). Houses represent land uses (e.g., disamenities or amenities).

and standard deviation for tracts in the 2010 United States Decennial Census. We use a gamma distribution because it is right skewed (as is true of housing values in the United States).

Residents

The key actors in the model are residents, who interact with their environment, which includes one another and characteristics of the land. The simulation begins with a population of 200 residents that grows at a 5% rate at each tick forward in time—the process is actually to eliminate 2% of the residents at each time tick and add 7% to keep the agents churning through the city in a dynamic process as time moves forward.[2] The time encompassed by a *tick* is fairly arbitrary, but in this context can be thought of as 1 year. Unlike Haase et al. (2010), we model a growing city since a city that is not growing is less likely to be adding residents and experiencing the housing shifts necessary for gentrification. Land uses (LUs) that are either amenities or disamenities are randomly placed at the start (see below), with new LUs added at the rate of one each per every 10 new residents.

Residential Attributes (Kind and Wealth)

Residents are intended to be simple but heterogeneous representations of people. They function according to only one goal—to find a location to live—and have two key attributes: *kind* and *wealth*. Kind is a variation on ethnicity or race. We use kind rather than these more familiar terms because our model does not include any historical discriminatory behavior or social distinction—kind represents a simple, mathematical majority/minority distinction, rather than a more pervasive minoritization process that characterizes racial and ethnic social relations. However, some discrimination can be included through the similarity preference function, discussed below. At the start of the model, 70% of the population is majority and 30% is minority. The population growth function randomly replicates the kind proportion of the population; thus, whereas the proportional breakdown can vary from this exact state, the proportions remain relatively consistent around this point.

To mirror the American context of our study, we assume that the majority is wealthier, on average, than the minority. Each time a new resident enters the model, it draws a wealth value from one of two gamma distributions, with the majority income distribution centered at $54,000 (the median family income of White Americans in 2010) and the minority income distribution centered at $32,000 (the median family income of African Americans in 2010), with both distributions defined by a standard deviation of $40,000 (Campbell et al., 2015).[3] As the simulation moves forward in time, residents' wealth levels change based on real estate transactions. This is done to account for the potential monetary benefit that homeowners in gentrifying areas may see when they sell their home for more than they paid for it. When residents move, their wealth is increased by the difference between what they *paid* for the home and what they *sold* the home for (or vice versa—wealth can decrease if they *sell* for less than they *paid*).

Residental Location Decision-Making

New residents, including those present at the initialization of the model and existing residents that are considering relocating, select a location using a basic utility function, subject to two constraints. Residents begin the location process with a set of plots that are available (i.e., those that are not roads and do not contain nonresidential LUs). Residents first satisfy the two constraints and then locate based on the plot of land that provides them with the highest utility within their constrained set of plots. For the first constraint, we assume that residents will not locate to a plot that they cannot afford, nor to one that is substantially cheaper than they can afford. We operationalize this in the model by assuming that residents will not select a plot that costs more than 3 times greater than their individual wealth level, a common recommendation in real estate purchases, nor a plot

that is less than their wealth. If no plots are available that fit this constraint, residents will remain at their current location.

For the second constraint, residents are also constrained by a similarity preference—which we model as an implementation of Schelling's (1971) insights, insights that have also been shown to be important when considering environmental justice outcomes (Campbell et al., 2015). In line with Schelling, we assume that residents will limit their choice-set of locations to ensure that some proportion (varied in each scenario) of other residents near them is of the same kind as themselves, and (as Schelling did) we model this as a constraint on location choice. The similarity preference can be thought of as an articulation of the level of discrimination via housing choice. Given previous research illustrating how important this similarity preference may be for location choices (Campbell et al., 2015), we also vary the similarity preference scenarios as described below. Higher or lower similarity preferences lead to cities that are more or less segregated.

In the model, the similarity preference functions as a constraint—residents will only calculate utility scores of plots that fit their similarity preferences (of having 30%, 50%, or 80% of the residents on contiguous plots be of the same kind as themselves). If no plots are available that fit this constraint, residents will either remain at their current location, or, if they are new to the region, will relax the similarity preference and select a plot that fits their price constraint and maximizes their utility.

Once these constraints are satisfied, residents select the plot in their constrained set that provides them with the highest individual utility. The utility function is based on how desirable any particular plot in the region is to each heterogeneous agent. The utility function balances each agent's distinct preferences with respect to price, quality, distance to amenities and disamenities, access to transportation routes, and density as arguments.[4] All preferences are unique to each individual agent—each agent is randomly assigned a price, quality, distance to LUs, access to transportation, and density preference from a normal distribution. That is, some residents may have a high concern about price, whereas others may be less concerned about price and more concerned about transportation access, for example. The price, quality, distance to LUs, and access to transportation distributions are the same for all simulations. In some cases we vary the mean for the density preference distribution to simulate different types of cities, at means of 1 resident per plot, or 5 or 10 residents per plot. Thus, each agent's individual location preferences are variable, but they are still clustered around a common point.

When residents select a location, they consider their individual preferences and compare each plot to see which one maximizes their preference structure. The utility function is shown in Equation (1):

$$u_{i,t} = p_{i,t}^{-\alpha} \cdot q_{i,t}^{\beta} \cdot l_{ik,t}^{-\gamma} \cdot r_{i,t}^{-\delta} \cdot d_{i,t}^{\varepsilon} \tag{1}$$

An individual agent's utility rate, u, for plot i at time t is a function of the price, p, of plot i at time t; the environmental quality, q; the shortest distance, l, to some LU, k; the shortest distance, r, to a transportation route; and the density value, d. In this function, we assume that residents would prefer low prices, high quality, to minimize distances to either amenity or disamenity LUs (we assume that all LUs offer some benefit to the resident, such as a place to work or spend leisure time), to minimize distances to transportation routes and thus minimize commute times, and a density level that is near their individual preference. The parameters α, β, γ, δ, and ε are weighted for each individual resident, assuming that residents have different LU preferences (Torrens & Nara, 2007).

After these constraints have been satisfied, all newly introduced residents move to the plot that has the highest utility score in their available choice set. *Existing* residents are mobile and may or may not relocate depending on a comparison between their current home and other plots in the region. As the simulation steps forward in time until the end (tick = 60), at each time tick 20% of the existing residents are randomly selected to consider relocating (Campbell et al., 2015). Those selected seek a location that fits their price and similarity constraints and scores the highest utility

value as described above. If they find a location with a higher utility value than their current location, they relocate. If there are no plots in their constrained choice set that increase utility above their current location, they do not move, with one exception. If the price of their current plot sufficiently exceeds their current wealth (as described in more detail below), then residents will move and we consider them to be *displaced*.

Displacement

Displacement in this sense can be conceived of in one of two ways: either homeowning residents are compelled to sell their home when its value becomes quite high relative to their wealth (e.g., because they can no longer afford property taxes), or, if residents are conceived of as renters, the rent exceeds their ability to pay and they are forced to relocate. We note that the consequences of these two options are different in reality since when owners sell they increase their wealth, but renters do not. As described below, we consider either description to illustrate possible displacement, even if the former might be considered utility enhancing whereas the latter is clearly utility reducing. Residents are displaced whenever the price of a plot is at least double the resident's current wealth. When this is the case, the resident relocates to a different plot that fits their price and similarity constraints. The proportion of residents that actually move at each tick forward varies as space becomes constrained by a growing population over time, so in early time periods the proportion of residents that relocates is close to 20%, whereas in later time periods this decreases to about 10%. This seems realistic, as populations tend to be more prone to relocate whenever there is ample space and some perceived risk of remaining in place (Kirschenbaum, 1996). In the model, as the city matures, the quality levels stablilize, reducing the incentive of residents to relocate. Through the trials, displacement occurs at lower rates than the rate of random relocation, but, as shown below, is prevalent.

Land Uses

LUs can be conceptualized as any nonresidential land use, and we distinguish between amenities and disamenities in a general sense. Amenities might be nonpolluting firms, parks, schools, commercial districts, or other LUs that residents might prefer to locate near. Disamenities might be polluting industries, polluted plots of land, transportation centers, jails, or any other LU that residents would prefer to live away from (if they don't work there) but that still offer some potential value. LUs are randomly assigned an amenity score ranging from 0 to 9. A score of 0 indicates a pure amenity, and 9 represents a pure disamenity, with scores of 0–4 indicating that an LU is on balance an amenity and scores of 5–9 indicating a disamenity. The lower the amenity score, the more residents seek to be located nearby. These amenity score levels are drawn from a uniform distribution. Thus, our assumption is that there are comparatively few pure amenities or disamenities, but rather a set of LUs that offer some benefits and some detriments. As mentioned above, LUs are added to the world at the rate of one for every 10 new residents. LUs select locations based on finding a plot that is not inhabited by any residents, and on minimizing price.

Land

Plots of land are modeled to function according to a real estate market, calibrated to function as realistically as possible given the model's parsimonious design.

Plot Price

Each plot's price is affected by residential location decisions nearby and the presence of LUs. First, plot prices are decreased based on the inverse distance to LUs. Basically, pure amenities with a score of 0 have no effect on plot price, but all other LUs have some negative effect on price—which may be indirectly balanced out by residential demand and resident preferences to locate

near amenities. Plot prices are also weighted at each tick by the average wealth of the residents located on the plot, and those on the eight contiguous plots (Veldkamp & Verburg, 2004), as shown in Equation (2):

$$p_{m,t} = \left[p_{m,t-1} - \left(p_{m,t-1}\left(0.01 * s_{k,t}\right)\right)\right] + \left[\frac{\sum w_{i,t}}{n_i} + \frac{\sum w_{j,t}}{n_j} - p_{m,t-1}\right] \quad (2)$$

In Equation (2), the price, p, of plot m at time t is a function of its price at time $t - 1$, decreased by 1% of the amenity score, s, of the nearest LU, k, at time t, balanced by the average wealth, w, of the residents, i, located on the plot, and by the average wealth of the residents, j, located on the eight contiguous plots (recall that more than one *house* can be on a plot under some density conditions). Basically, plots with wealthier residents nearby and closer to amenities cost more (and vice versa).

Plot Quality

Quality is adjusted similarly, although this value is not affected by residents and is only affected directly by the closest LU. Quality is adjusted up or down based on the amenity score of that nearby LU, as shown in Equations (3a) and (3b):

$$If\ k = 0 - 4:\ q_{m,t} = \left[q_{m,t-1} + \left(q_{m,t-1}\left(0.01 * \left(4 - s_{k,t}\right)\right)\right)\right] \quad (3a)$$

$$If\ k = 5 - 9:\ q_{m,t} = \left[q_{m,t-1} - \left(q_{m,t-1}\left(0.01 * s_{k,t}\right)\right)\right] \quad (3b)$$

In these equations the quality q for plot m at time t is increased as in Equation (3a), or decreased based on Equation (3b), depending on whether the closest LU, k, is an amenity or disamenity. Quality is then altered by comparing the focal plot's quality level with the average quality of the eight contiguous plots. If the average quality level of the contiguous plots is greater than the quality level of the focal plot, then the focal plot's quality is increased by 1% of the difference. Conversely, if the focal plot's quality is higher than the average of the contiguous plots, then its quality is decreased by 1% of the difference.

Defining Model Outcomes

Gentrification

The literature on gentrification does not offer clear guidance on when, precisely, a neighborhood becomes gentrified. Without clear empirical guidance, we operationalize it as a calibrated process. First, a plot must be gentrifiable—that is, it must reach a state where it declines enough to become a potential target for gentrification. We conceive of this simply: if a plot's price is in the bottom quartile of plot prices we consider it gentrifiable. We consider a plot to be gentrified only if it first becomes gentrifiable, and then subsequently sees a price increase into the upper two quartiles of price.[5] Once land is gentrified, it is no longer gentrifiable, unless it subsequently drops back into the lowest quartile of land values.

Displacement

We assess the amount of displacement from gentrified plots. Because residents are mobile and some move for different reasons, we assume that residents can only be displaced if they lived on (a) a gentrifiable plot that was (b) subsequently gentrified while they were still located there, and then (c) relocated to a different plot that was gentrifiable (but not gentrified at that time) because they met our housing burden threshold as described above. We believe this reasonably captures the type of displacement that is of concern to those worried about gentrification-related displacement. We track the residents that move, for this reason, from gentrified plots to gentrifiable plots, assessing these rates for different wealth and kind classes of residents.

Scenarios and Setup

To ensure that we begin with a region that has ample opportunity for gentrification and displacement, we begin each trial by engineering the region to have relatively high levels of segregation and disparities between the kinds of residents. Basically, we try to mirror an American urban setting, around the time we began to note gentrification occurring (mid to late 1980s) and assess results from this starting point. For the first 25 time ticks[6] of each scenario, we ensure that the majority group lives on plots with higher quality levels and that both groups locate with very high similarity preferences (each resident class prefers 80% of residents on the eight contiguous plots to be of the same kind). After tick 25, we alter the decision-making process for residents and begin a phased removal of disamenities to spur redevelopment and potential gentrification (Checker, 2005; Eckerd, 2011). At each tick after tick 25, we assess gentrification and displacement in a variety of different contexts and assess what happens to the residents involved.

Disamenity Removal Policy Scenarios

In all scenarios we implement a policy removing one existing disamenity from the region per time tick as our method to spur gentrification. As shown in Table 1, in some scenarios the disamenity that is on the highest priced plot is removed (illustrating an assumption that the plot has redevelopment potential), and in others the disamenity located near the largest clusters of minority residents is removed (illustrating a scenario of targeted cleanup of minority areas). We also vary what occurs after the disamenity is removed: in some cases, it is simply removed, whereas in others a disamenity is *redeveloped* into an amenity. We expect that when disamenities are removed the plot attributes will be affected by an increase in both quality and price, which will change how this plot is perceived through the residential utility function. We assume that removal of the disamenity will immediately change the plot's quality, akin to removal of a hazardous site, but the price change will lag behind, making the site attractive to new or relocating residents who can take advantage of low prices and comparatively higher quality. We expect both of these processes to quicken when the disamenity is immediately redeveloped into an amenity—quality will rise more quickly, and price will follow closely.

Residential Preference Scenarios

We vary the context in two ways: by changing similarity preferences and density preferences to account for variation in residents and their cities. Since each *city* we compare has the same geographic form, we varied similarity preferences to account for cities that vary in the extent to which there is high or low racial segregation in settlement patterns, and density preferences to account for cities that are generally more or less dense. The density preference could also be thought of as a geographic constraint. In some highly dense cities, residents may simply prefer more dense settlement or there may be geographic constraints (such as waterways or mountains)

Table 1. Scenarios and variable settings.

Variables	Distinct assumptions		
Disamenity priority	Clean up Disamenity on highest-priced plot	Clean up Disamenity on plot with highest density of minority residents nearby	
Disamenity process	Disamenity is removed; land is available for land uses and residents	Disamenity is removed and replaced with an amenity	
Similarity preference	No segregation: 0% for both majority and minority groups	Low segregation: 30% preference for both groups	High segregation: 80% majority preference, 50% minority preference
Density preference	Low density: Mean density preference of 1 resident per plot	Moderate density: Mean density preference of 5 residents per plot	High density: Mean density preference of 10 residents per plot

or policy constraints (such as urban growth boundaries) that enforce higher density. From our modeling perspective, the result is the same—in some cities residents live in dense urban settings, whereas in others there is primarily less dense, more suburban-style settlement. For our scenarios, we varied similarity preferences in three ways—two hypothetical situations (0%, 30%) and an empirical finding (50% minority/80% majority similarity preferences, Clark, 1992)—density preferences in three ways (low, medium, and high), disamenity removal priority in two ways (as described above), and disamenity removal processes in two ways (all as depicted in Table 1). This results in a total of 36 distinct scenarios ($3 \times 3 \times 2 \times 2$). We ran each scenario for 100 trials, each with 60 ticks per trial,[7] and aggregated the means of all key measures for the trials for each scenario.

Results

Policy Choices

In our initial analysis of the results, it was clear that some of our experimental variables had very little effect on the outcomes of focus. Whether disamenities were cleaned up or cleaned up and replaced with amenities had virtually no effect on gentrification or displacement outcomes. Further, results were also indistinguishable whether cleanup was targeted to spur economic development by remediating the highest priced plot or to improve environmental equity by remediating disamenites in minority areas. These findings alone are important, for they imply that *where* we choose to clean up (on high-value lands or in minority communities) and *whether* we choose to immediately redevelop undesirable land or facilities are not as important as understanding local preferences around similarity and density (recalling that the latter can include transportation efficiency).

Residential Preferences

The differences we encountered by varying the residential preference variables—similarity preferences (segregation) and density preferences—far overwhelmed the minor differences for the disamenity policy choices. Therefore, for simplicity in presenting our results, we condensed the 36 scenarios to nine, varying only segregation and density preferences. We did so by averaging the outcomes of all trials across the disamenity policy choices into just the nine scenarios shown in Table 2. In the figures we depict the time trends of the key variables, showing the lines only when the results are statistically significant (at the 95% confidence level), including aspects of the land and residential outcomes. Graphs are labeled with the scenario numbers shown in Table 2.

Gentrifiable and Gentrified Land

We note that in all scenarios gentrifiable land was prevalent. This is not surprising because, by our definition, at each tick 25% of all land is gentrifiable. As shown in Figure 2, in each scenario about 40% of land was, at some point over the course of the 60 ticks of the scenario, classified as gentrifiable, and between 20% and 35% of all land was subsequently gentrified.

There was consistently more gentrification when the residents had higher density preferences and, to a lesser extent, when they had higher similarity preferences. In low-density preference

Table 2. Scenario guide.

	Segregation: None	Segregation: Low	Segregation: High
Density: Low	1	2	3
Density: Mid	4	5	6
Density: High	7	8	9

Note. In reporting results for these nine scenarios, results are averaged over the minor differences caused by differences in disamenity priority and disamenity process.

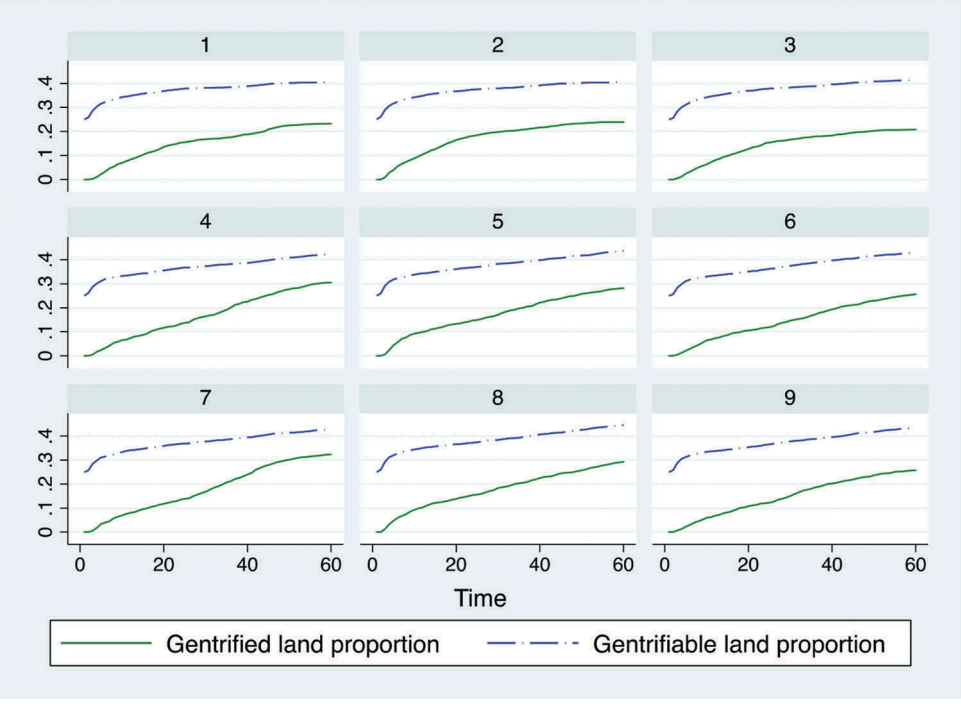


Figure 2. Cumulative gentrifiable and gentrified land proportions.

scenarios (scenarios 1, 2, and 3), only about 20% of the land gentrified, whereas around 30% did in higher density preference settings. Land was more likely to gentrify in scenarios with moderate or high density preferences and no or low segregation preferences (Scenarios 4, 5, 7, and 8). The least amount of gentrified land for each density assumption took place in the highly segregated settings (scenarios 3, 6, and 9). Nevertheless, as Figure 2 shows, in all cases, by the 60th tick between 20% and 30% of all land moved from the lowest quartile (gentrifiable) to the upper two quartiles (gentrified), showing that there was clear movement of comparatively wealthy residents into comparatively poor areas. But in no case did all the gentrifiable land gentrify. In short, Figure 2 basically matches some things that we know empirically and provides some validation that our model is consistent with observations—gentrification occurs, but not all poor areas of a city will gentrify.

Displacement

In Figure 3, we show the proportion of residents who were displaced at least once from gentrified neighborhoods, looking first only at the displacement of minority-kind residents, because one concern around gentrification is racial displacement. The minority displacement rate was highly influenced by the density preference. In low density preference settings (scenarios 1, 2, and 3), about 25–35% of minority residents were displaced at some point during the trial. At higher density preference settings, about 60–80% of minority residents were displaced at least once, with the displacement rate being *inversely* proportional to segregation preferences. About 80% of minority residents were displaced at least once in scenarios with a preference for moderate or high density but no segregation preference (scenarios 4 and 7), and only about 60% in scenarios with high levels of segregation (scenario 9, for example).

We also compare displacement rates for minority and majority residents, as shown in Figure 4. Note that lines are shown in Figure 4 only when differences between displacement rates for

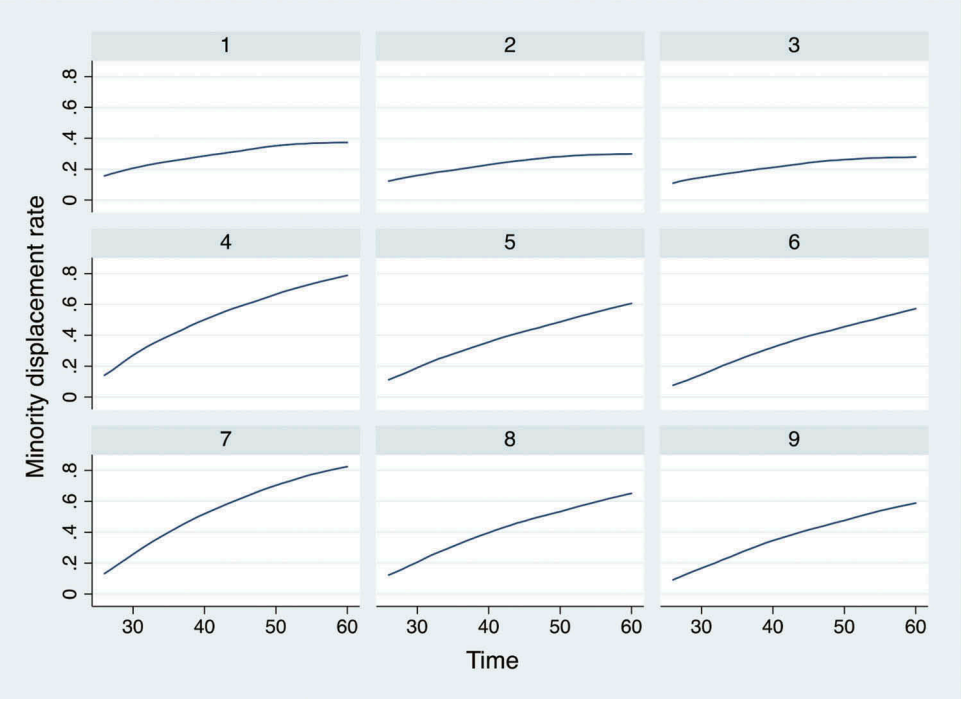


Figure 3. Minority displacement rates over time.

Note. All results are statistically significantly different from zero with 95% confidence.

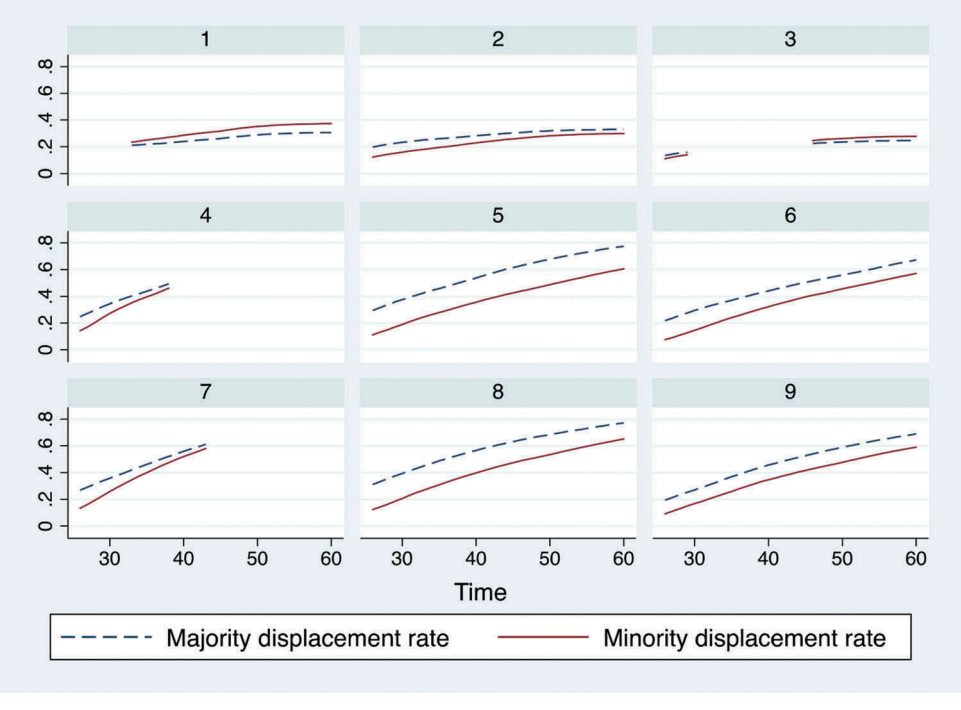


Figure 4. Comparison of majority and minority displacement rates.

Note: Lines are only shown when differences are statistically significant with 95% confidence.

majorities and minorities are statistically significantly different (at the 95% confidence level). Majorities are generally displaced at higher rates than minorities are in most scenarios, but the differences illustrate important distinctions. First, minority residents are only displaced at higher rates than majorities in two of the low density preference settings (1 and 3), although the rate of displacement is quite close under all of the scenarios with no segregation preference (scenarios 1, 4, and 7). Majorities are displaced at the highest rates when there is a medium or high density preference, and are displaced at higher rates than minorities when there is also a moderate to high segregation preference (scenarios 5, 6, 8, and 9).

Figure 4 does not differentiate residents by wealth class. If we narrow to considering only the residents in the poorest quartile of both kinds, as shown in Figure 5, the conclusion is the same: we find there is little displacement of the poor when residents prefer low density (Scenarios 1, 2, and 3). Once there is a preference for density, the minority poor do worse than the majority poor in conditions where either there is no preference for segregation (4 and 7), or the preference for segregation is quite high, as in scenarios 6 and 9 (closer to what we observe empirically in the United States).

The proportion of residents displaced at least once provides some insight into the burdens of gentrification-related displacement, but a fuller understanding of the burden of displacement in a dynamic housing market can be considered by the total number of times that poor residents were displaced, as shown in Figure 6. First, poor residents are displaced very infrequently in low density preference contexts, as seen in Figure 6, scenarios 1, 2, and 3 where the trend line is barely above zero. Whereas about 35% of poor residents are displaced at least once in these scenarios (see Figure 5), very few are displaced more than once. Conversely, in contexts where there is higher density, residents can be displaced multiple times, and as the city grows the number of times displaced seems to grow exponentially. In situations with low segregation preferences and higher

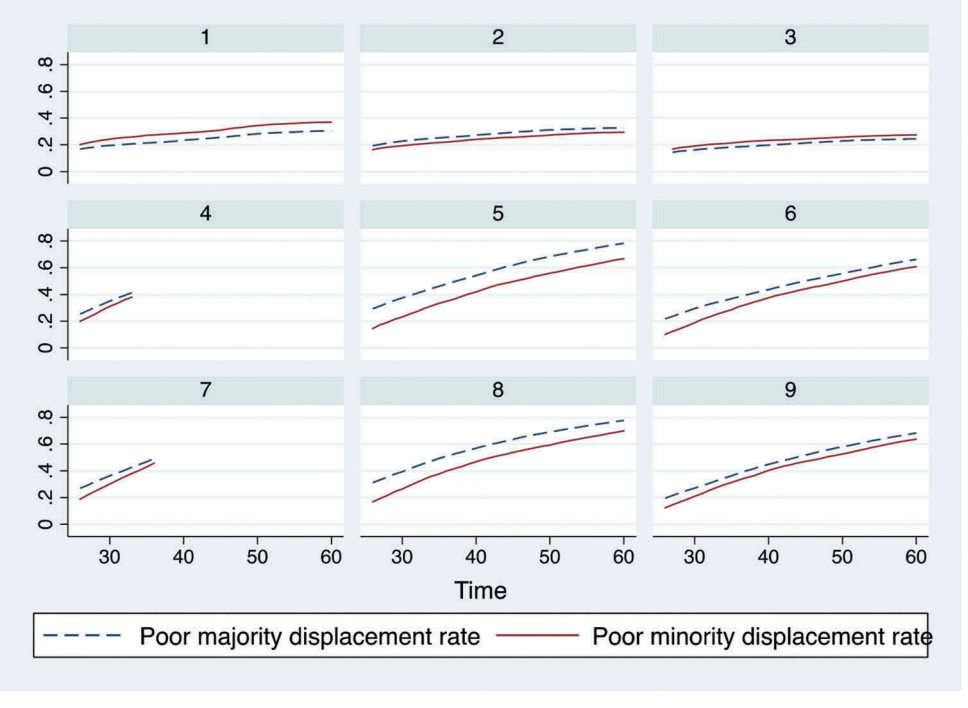


Figure 5. Comparison of poor majority and poor minority displacement rates.

Note. Lines are only shown when differences are statistically significant with 95% confidence.

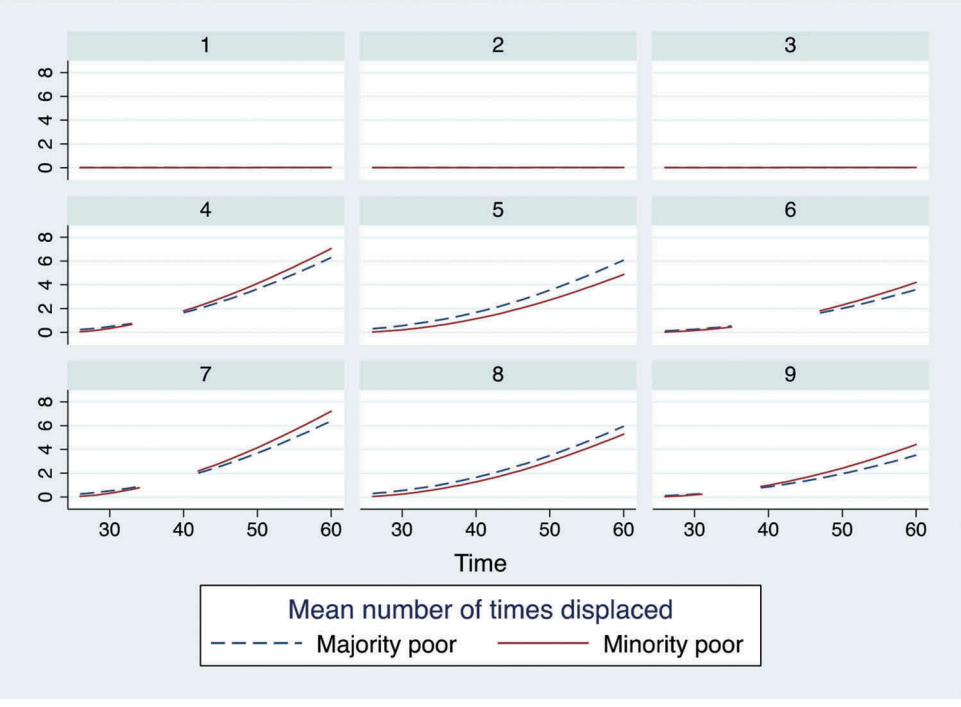


Figure 6. Mean number of times poor majority and poor minority residents were displaced.

Note: Vertical axes measure counts (rather than proportions as in Figures 2–5). Lines are only shown when differences are statistically significant with 95% confidence.

density preferences (scenarios 4 and 7), the average number of times a poor resident is displaced during the time period of the model is six. For context, since we are assessing gentrification over 35 ticks, or the equivalent of a 35-year period, this means that poor residents in these settings were displaced about once every 6 years. Residents of both kinds are displaced more frequently when residential segregation preferences are low and less frequently when segregation preferences are high (compare 4 with 5 and 6, and 7 with 8 and 9). Poor minority residents are displaced slightly more often than majority residents are, except in the moderate segregation setting (scenarios 5 and 8).

Discussion and Policy Implications

Although these results should be understood in the context of a stylized simulation model, our model outcomes offer some useful insight for understanding the gentrification and displacement processes and how policy may be able to address these issues. Or, more to the point, our results suggest that the same policies may result in different outcomes depending on the urban context in which the policy is being implemented, for the processes may be spurred much more by structural aspects of an urban region than by a policy response; this matches findings by Fitzgerald and Vitello (2014) when looking empirically at effects of the Community Reinvestment Act. Concerns have been raised about the potential for economic development policy to spur displacement (Checker, 2005), but this may be true only in a relatively small number of urban contexts: those with high levels of density preferences and residents who are relatively open to living in proximity to people unlike themselves. In lower density preference contexts, or where there are high levels of segregation preferences by residents, displacement seems to be a less common experience.

Our analysis suggests that appropriate policy responses to addressing economic redevelopment through environmental improvement may not be a one-size-fits-all proposition. One of the useful features of an ABM is coming up with unexpected outcomes that emerge because of the interactive features of the modeled process. We certainly did not expect residents' density preferences to overwhelm the effect of LU policy considerations. Yet, in low density preference areas, disamenities were removed without spurring high levels of displacement. This suggests that cities with low density may have flexibility in how they address disparities and may be able to relax concerns about widespread displacement, because even though some displacement did occur, neighborhoods did not tend to change as much as they did during the higher density preference trials. In contrast, in higher density preference areas, particularly those with no preferences for segregation, displacement affected almost all poor residents (up to 80% in some cases) and residents were displaced multiple times—a true hardship, especially for the poor. In cities fitting these profiles, policymakers may need to address place-based economic problems and environmental concerns much more cautiously. Perhaps unexpectedly, this advice may apply less to areas with high levels of segregation preferences, as gentrification and displacement were less likely in these areas. Although the social problems of *segregation* remain, the *displacement* problem appears less likely in high-segregation settings.

Stability in low-density areas is not a new finding (Mieszkowki & Mills, 1993), and our results here suggest that when residents prefer low density, stability will be high and the benefits of redevelopment will tend to flow to all residents who live in any particular area (rather than to incoming gentrifiers). In contrast, with high density levels, residents are more mobile and neighborhoods are characterized by less stability. This is particularly true when residents are comfortable living around individuals different from themselves, for in these cases segregatory preferences do not provide barriers to movement. However, when residents hold high similarity preferences (i.e., they exhibit high levels of racial discrimination in housing), neighborhoods are more stable because individuals are less willing to relocate to areas with residents who are different from themselves.

Policy Implications

As noted above, our results provide less in terms of policy-specific guidelines but more in terms of useful suggestions regarding the context of the policy approach. Our model suggests that environmental cleanup policy should be considered within the setting of the structural characteristics of the urban setting, with different policy recommenations for different types of cities. In Table 3, we propose an urban context typology that could be used to consider the probability and potential burden of gentrification-based displacement.

We consider some examples to illustrate this typology.

High Density, Low Segregation Preferences

New York City may represent an area with high density and low segregation preferences (at least today, if not historically). Although the city is increasingly segregated by income class, a low segregation preference by race/ethnicity and high density preference may explain the high rates of gentrification and displacement that have been occurring there for decades (Lees, 2003).

Table 3. Gentrification context probability and burden on displacees.

	Segregation preferences	
Density	High	Low
High	High probability of displacement with moderate burden on displacees	High probability of displacement with high burden on displacees
Low	Low probability of displacement with low burden on displacees	Low probability of displacement with moderate burden on displacees

High Density, High Segregation Preferences

Chicago, Illinois, may represent a high density/high segregation city, as gentrification has certainly occurred, but the well-known racial distinction between the city's north and south sides has remained stable for decades (Silver, 2015).

Low Density, High Segregation Preferences

Cities with longstanding segregation but low density may tend to have stable neighborhoods with little gentrification or displacement, or gentrification that is very slow to take hold, as is the case in many smaller cities with stable economies, including cities with good economies like Omaha, Nebraska (Gonzalez, 2016), and poor economies like Stockton, California (Runner, 2016), or places in between like Indianapolis, Indiana (Taylor, 2016).

Low Density, Low Segregation Preferences

Finally, low-density cities with lower (or changing) segregatory preferences, like Charleston, South Carolina (Faussett, 2015), or Durham, North Carolina, may begin to see more, albeit slowly developing, gentrification and related displacement (White, 2016).

Overall

Each of these different settings may require a different policy approach to limit or compensate for displacement. Low-density cities, particularly those that are highly segregated, may not need to be very concerned with displacement as it may be less prone to occur there. This leaves them free to target removal of disamenities in minority neighborhoods.[8] Higher density cities, particularly those with comparatively low levels of segregation, may need to consider policies well beyond limitations on disamenity siting or targeted disamenity removal. Although our model did not assess specific housing policies, it suggests that in dense, diverse cities, displacement may be a significant problem.

Given the very limited effect that LU policy differences had on our outcomes (gentrification and displacement), policy in these cities may need to focus on finding a balance that provides needed economic improvements to certain neighborhoods, but perhaps not improvement that would appeal to a broad population; or perhaps improvements need to be balanced with other policies. For example, zoning or use restrictions, or even easements or other modes of land conservation, might be useful.

These contextualized results, which suggest that policy results depend on the context and not on a single *all else being equal* outcome that can be applied to all U.S. cities, are the sort of results that suggest the importance and value of the ABM approach, with its focus on the interactive realities of complexity. Nonetheless, our results, and these suggested policy implications, should be considered in the context of the limitations of ABM as a method. First, the ABM is a highly stylized and simplified model. Its usefulness depends on the validity of the representation of agents, their interaction rules, and their environmental conditions. For any ABM, the appropriate level of abstraction and simplification is largely dependent upon the purpose of the model, and also upon our current understanding of the process. Although we built the model based on our best knowledge, and tested its sensitivity to changes in parameters (see the appendix), the assumptions are open to debate. Nevertheless, the goal of building an abstract model of gentrification and displacement is to focus on and experiment with plausible social processes, to identify the *social recipe* that leads to certain aggregate outcomes—not to estimate the effect sizes of specific parameters as in regression analysis. In fact, since the model relaxes the typical statistical assumption of the independence of input variables, standard effect-size considerations are not applicable. The model assumes the interdependence of all input variables, and thus any particular scenario result is an outcome of an interaction between variables. Any assessment of the *independent* effects of these interdependent variables would provide a false picture of the nature of the model. Although stylized, the formal dynamism of the ABM creates an understanding of social processes

that can lead to certain outcomes, allowing us to craft more context-specific policies that take into account the complex dynamics of the social world, and less content-specific policies that assume the indepence of variables that, we argue, are not independent.

Although our model suggests that gentrification and policies to address displacement disparities probably ought to be considered within the context of segregation and density trends of a city, we do not suggest that these considerations should be the only considerations. Nevertheless, being able to describe our results in reference to real cities illustrates the potential utility of the model's insights.

A final limitation is our operationalization of racial distinctions. In our model, difference is a relatively benign concept. It is a distinction of mathematical majority/minority, with some discriminatory preference expressed through the residential similarity preference. With simple agents that possess no social memory, and with simplistic behavioral assumptions, we do not model anything as complex as actual racial disparity and segregation—the agents have no institutional memory of historical racism or contemporary discrimination—although the accretion of these realities may be instantiated in a very simple way via the similarity preference.

Conclusion

In this research, we model gentrification and gentrification-related displacement of residents. We do so through the development of an ABM of a simple urban region, considering different urban contexts and policy approaches to improving environmental conditions and the relationship of these policies to subsequent gentrification and displacement. We find that gentrification-related displacement is most likely, and most potentially burdensome, in urban regions characterized by high levels of density and low levels of residential segregation preferences. Displacement is far less prevalent in low-density regions, particularly those with high segregation preferences. We find that choosing to clean up disamenities based on land value or minority area makes little difference to these outcomes, as does whether the disamenity is simply removed (and the land is thus available in the market) or is removed and immediately replaced with an amenity. Our model indicates that the characteristics of the region may affect the policy outcome more than any particular policy approach does. In areas with high displacement, any policy to improve environmental conditions fostered environmental improvement that disproportionately benefited incomers, whereas in low-displacement regions any policy tended to effectively address disparities in living conditions. Although disamenity cleanup may be well intentioned, its impacts for residents appear to depend on the local context, and decision-makers should take this into account before acting.

Notes

1. When residents have higher density preferences, modeled cities have higher density housing outcomes.
2. These eliminations can be thought of as people moving completely out of this city to another, or also as losses because of death, etc.
3. The gamma distribution is used because it is right skewed, as is the U.S. income distribution. Both distributions are constrained to be no less than zero. Data for the income variable was derived by the authors from the 2010 U.S. Census. Using the Census block level as the unit of analysis, we imputed a data set of individuals in the United States via the population of each census block, a random assignment of individual incomes based on the average income in each census block.
4. We thank an anonymous reviewer for suggestions about the utility function and the modeling strategy generally. After conducting sensitivity analyses, the model was altered to better capture the heterogeneity of agent decision-making. See the Appendix for more detail about the model.
5. As a first attempt to operationalize this concept in a simulation model, we arrived at this operationalization through testing. We set lower thresholds for *gentrifiable* and higher thresholds for *gentrified*, settling on these values as a good balance to ensure that some gentrification took place in most trials but did not overwhelm

the entirety of the region. This operationalization also seems to fit common-sense notions of what it means for a neighborhood to be gentrified.

6. For simplicity, we conceive of each time tick as 1 year, but the time ticks may represent any relevant time designation.
7. Although we could have run each simulation longer, model testing revealed that price and quality trends stabilized at about tick 60, so we cut off the trials at this point; if thinking in terms of years, this suggests that we began the models around 1980 and concluded them at roughly 2020. See the appendix for more information.
8. Note that the finding that gentrification displacement is not affected by policy decisions regarding cleanup does not imply that other outcomes, such as environmental justice, are not affected by them.

Disclosure Statement

No potential conflict of interest was reported by the authors.

Notes on Contributors

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Appendix. Model robustness assessments.

Parameters	Model value	Range of values tested	Result
Land uses			
Number of amenities/disamenities per resident	10, fixed	10–50 by intervals of 10	At higher numbers, fewer firms were established, thus less quality variability and generally less gentrification. Ten provides a good balance between a reasonable assumption and enough variation in quality and resident proximity to land uses. Results at high levels showed few discernible trends, but as the number of residents per land use was lowered, similar patterns to those presented emerged.
Residential proximity preference	0–10, uniform distribution	0–10, from normal distribution ~ (5, 2)	Normal distribution resulted in many firms near 5, and little variation in residential proximity and its effects. Results were similar but muted.
Location preference/cleanup priority	Lowest price; near minority residents	Random, low price, near minority residents, away from majority residents	Since the minority population was assumed to be less wealthy than the majority, model results were robust to any location preference or cleanup priority for land uses.
Residents			
Similarity preference	0, 0.20, 0.50/0.80 as a location constraint	0.20, 0.50, 0.80; similarity preference part of utility calculation rather than constraint	Results with similarity preference in the utility function were conceptually the same as those presented, with a dampened effect. It was left as a constraint to remain consistent with Schelling (1971).
Utility preferences	Normal distribution assignment; equal weighting in utility function; price, quality, density, and transportation access	Combinations of different weighting; removal and addition of some preferences; including density as a constraint rather than a preference.	Different weighting of the preferences had little effect on results. Density as a constraint enhanced the effect of density, but the literature suggested a preference in the utility function as more justified.
Patches			
Transportation	Proximity to roads as a resident preference	Proximity to roads not a resident preference	A preference to be located near transportation routes muted the effects of land-use changes on gentrification, but enhanced them near transportation routes. The preference was included as it is realistic to assume that areas with transportation access would be prone to gentrification.
Price	Random draw from gamma distribution of U.S. Census values	Random draw from normal distribution	The distribution type made little difference in model results. Gamma distribution was used as the more justifiable option.
Price and quality diffusion	Based on proximity to LUs and wealth of nearby populations	Based on quality; diffusion of prices and quality from nearby plots	Reliance of quality as a price mechanism quickly made quality patches cost too much for even wealthy residents. Proximity to LUs and wealth led to somewhat stable but not completely stable neighborhoods, which was determined to be realistically valid.

(*Continued*)

Appendix. (Continued).

Parameters	Model value	Range of values tested	Result
Global			
Ticks	60	Range: 50–200 by intervals of 25	Although somewhat variable, between about 50 and 100, the trends presented tended to continue on similar trajectories after tick 60.
Period before policies implemented	25 ticks	Range: 10–70 ticks, by intervals of 10	Environmental justice problems only became more significant if more ticks passed before policy implementation. Tick 25 seemed a good choice, as trends were clear but not yet overwhelming.
World size	40 × 40	10 × 10 to 100 × 100 in intervals of 10	Trends were similar regardless of world size. Emergence of gentrification was faster in smaller worlds and slower in larger worlds. A 40 × 40 world was a good balance, with a sizable city that did not overtax available computing resources.
Population growth rate	5%	0–25%	Results were similar regardless of population growth rate, although at higher growth rates space became more constrained, so initially the trends were amplified in both the set-up and policy implementation phases, with trends leveling out earlier than 60 ticks.